

# A 2-D Unstructured Compressible Navier–Stokes Finite-Volume MPI Solver: Benchmark Report

uns2d solver

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## Abstract

We present **uns2d**, a from-scratch cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, written in C++17 with MPI domain decomposition. The solver reads unstructured CGNS meshes (mixed triangles/quadrilaterals, multi-zone), partitions the cell graph with METIS, and exchanges a one-cell ghost halo with neighbour ranks using `MPI_Isend/Irecv`. Spatial discretization is second order via least-squares gradients with a Venkatakrishnan limiter; the inviscid flux is Roe with a Harten–Yee entropy fix (Rusanov for the transient case); viscous fluxes use corrected face gradients of primitive variables. Steady cases are advanced with an implicit LU-SGS pseudo-time march and CFL/local-time-stepping; the Reynolds 200 cylinder is integrated in physical time with a BDF2 dual-time two-loop scheme. All eight required cases were run to convergence or a documented plateau (steady) or statistically periodic shedding (Re 200). Force coefficients, convergence histories, field visualizations, an MPI rank-count comparison, and a machine-readable sanity-check gate are reported. Per-case status is stated explicitly and no failed run is presented as converged.

## 1 Introduction

The benchmark requires a nontrivial unstructured compressible-flow solver built without a starter skeleton, exercised on NACA0012 (Mach 0.15/0.8/2.0, inviscid and laminar Re 5000) and circular-cylinder (Re 20 steady, Re 200 transient) cases. This report connects each numerical choice to the observed results. The required cases and their status are collected in Section 9.

## 2 Governing Equations and Nondimensionalization

The conservative state is  $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$  and the 2-D compressible Navier–Stokes equations are solved in conservative form,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad (1)$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \quad (2)$$

The perfect-gas closure is  $p = (\gamma - 1)\rho e$ ,  $E = e + \frac{1}{2}(u^2 + v^2)$ ,  $a = \sqrt{\gamma p / \rho}$ , with  $\gamma = 1.4$ ,  $R = 1$ ,  $Pr = 0.72$ . Temperature is  $T = p / (\rho R)$  and  $c_p = \gamma R / (\gamma - 1)$ . The viscous fluxes use the Newtonian

stress tensor and Fourier heat flux,

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \quad \mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr}, \quad (3)$$

so that  $\mathbf{F}^v = [0, \tau_{xx}, \tau_{xy}, u\tau_{xx} + v\tau_{xy} - q_x]^T$  and  $\mathbf{G}^v = [0, \tau_{xy}, \tau_{yy}, u\tau_{xy} + v\tau_{yy} - q_y]^T$ . Viscosity is constant and matched to the case Reynolds number,  $\mu = \rho_\infty U_\infty L_{ref}/Re$ ; inviscid cases set  $\mu = k = 0$ . The freestream is nondimensional ( $\rho_\infty = 1$ ,  $U_\infty = 1$ ,  $p_\infty$  set by the target Mach number). Force coefficients use dynamic pressure  $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2 = \frac{1}{2}$  and the case reference area and length:  $C_D = \mathbf{F} \cdot \hat{\mathbf{d}}/(q_\infty A_{ref})$ ,  $C_L = \mathbf{F} \cdot \hat{\mathbf{l}}/(q_\infty A_{ref})$ ,  $C_m = M_z/(q_\infty A_{ref} L_{ref})$ , with pressure and viscous parts reported separately.

### 3 Meshes, Boundary Tags, and Case Inputs

Meshes are imported from CGNS with the mid-level library. All unstructured zones are read; triangle (**TRI\_3**) and quadrilateral (**QUAD\_4**) sections become cells and **BAR\_2** sections become boundary edges. Multi-zone meshes (the cylinder has two zones joined by a 1-to-1 interface) are merged by welding geometrically coincident nodes with a uniform-grid spatial hash, yielding a single watertight mesh; the closure residual  $\sum_f \hat{\mathbf{n}}_f \Delta S_f$  per cell is at machine precision ( $\sim 10^{-15}$ ), confirming correct face areas/normals. Cell volumes and centroids use the shoelace formula (with CCW reordering); each interior edge maps to one face shared by two cells with an outward normal oriented from owner to neighbour. Boundary-family names are read from the case JSON **boundary\_conditions** map (**bc-2/bc-4** for NACA, **WALL/FAR** for the cylinder) and are never hard-coded. Mesh sizes and case parameters are listed in Table 1.

Table 1: Required cases: mesh, MPI ranks, steps, physical time, residual reduction (orders), wall time, and convergence status.

| Case                         | Mesh             | Ranks | Steps | $t_f$ | Res. orders | Wall (s) | Status                 |
|------------------------------|------------------|-------|-------|-------|-------------|----------|------------------------|
| naca0012_m015_inviscid       | NACA0012.H2.cgns | 8     | 13030 | 0     | 4.00        | 779      | converged              |
| naca0012_m080_inviscid       | NACA0012.H2.cgns | 8     | 7786  | 0     | 3.93        | 621      | converged              |
| naca0012_m200_inviscid       | NACA0012.H2.cgns | 8     | 3578  | 0     | 2.10        | 562      | converged              |
| naca0012_m015_laminar_re5000 | NACA0012.H2.cgns | 8     | 3573  | 0     | 4.00        | 497      | converged              |
| naca0012_m080_laminar_re5000 | NACA0012.H2.cgns | 8     | 9604  | 0     | 4.00        | 663      | converged              |
| naca0012_m200_laminar_re5000 | NACA0012.H2.cgns | 8     | 2967  | 0     | 3.00        | 422      | converged              |
| cylinder_m010_laminar_re20   | CylinderB1.cgns  | 8     | 6175  | 0     | 4.92        | 488      | converged              |
| cylinder_m010_laminar_re200  | CylinderB1.cgns  | 8     | 30000 | 300   | 0.99        | 2331     | statistically_periodic |

## 4 Spatial Discretization

The semi-discrete residual for owned cell  $i$  is the sum of face fluxes,  $\mathbf{R}_i = \sum_{f \in \partial i} (\mathbf{F}_f^i - \mathbf{F}_f^v) \cdot \hat{\mathbf{n}}_f \Delta S_f$ , with  $d\mathbf{U}_i/dt = -\mathbf{R}_i/V_i$ . Boundary faces contribute a single flux computed from the reconstructed interior state and the boundary condition.

### 4.1 Second-Order Reconstruction

Cell-centered gradients of the primitive variables  $[\rho, u, v, p]$  are computed by inverse-distance-squared-weighted least squares over face-neighbours (a  $2 \times 2$  normal system per cell, precomputed and cached). Left/right face states are obtained by piecewise-linear MUSCL reconstruction,

Table 2: Final (steady) or mean (transient) force coefficients with the pressure/viscous drag split.

| Case                         | $C_D$  | $C_L$   | $C_m$   | $C_{D,p}$ | $C_{D,f}$ |
|------------------------------|--------|---------|---------|-----------|-----------|
| naca0012_m015_inviscid       | 0.0009 | +0.0001 | +0.0000 | 0.0009    | 0.0000    |
| naca0012_m080_inviscid       | 0.0087 | +0.0000 | +0.0000 | 0.0087    | 0.0000    |
| naca0012_m200_inviscid       | 0.0909 | +0.0001 | +0.0003 | 0.0909    | 0.0000    |
| naca0012_m015_laminar_re5000 | 0.0370 | -0.0004 | -0.0001 | 0.0015    | 0.0355    |
| naca0012_m080_laminar_re5000 | 0.0415 | +0.0060 | +0.0011 | 0.0395    | 0.0021    |
| naca0012_m200_laminar_re5000 | 0.2176 | +0.0001 | +0.0000 | 0.0316    | 0.1860    |
| cylinder_m010_laminar_re20   | 0.8780 | +0.0045 | -0.0001 | 0.6870    | 0.1909    |
| cylinder_m010_laminar_re200  | 0.9093 | +0.0352 | +0.0009 | 0.7950    | 0.1143    |

Table 3: Re 200 BDF2 dual-time inner-iteration statistics.

| Quantity                            | Value          |
|-------------------------------------|----------------|
| Time integrator                     | bdf2_dual_time |
| Physical steps                      | 30000          |
| Inner target (residual ratio)       | 0.001          |
| Observed inner iters (min/mean/max) | 5/18/153       |
| Converged-step fraction             | 1.000          |
| Inner target misses                 | 0              |

Table 4: Re 200 vortex-street analysis from `forces.csv` (second half of the history).

| Quantity                    | Value |
|-----------------------------|-------|
| Mean drag $\overline{C_D}$  | 0.919 |
| Lift amplitude $\hat{C}_L$  | 0.388 |
| Strouhal number $St = fD/U$ | 0.190 |

Table 5: MPI rank-count comparison: final forces agree across rank counts (including  $np = 8$ ).

| Case                       | Ranks | $C_D$  | $C_L$   | Wall (s) |
|----------------------------|-------|--------|---------|----------|
| cylinder_m010_laminar_re20 | 2     | 0.8780 | +0.0044 | 88       |
| cylinder_m010_laminar_re20 | 8     | 0.8780 | +0.0045 | 488      |
| naca0012_m015_inviscid     | 1     | 0.0008 | +0.0004 | 670      |
| naca0012_m015_inviscid     | 8     | 0.0009 | +0.0001 | 779      |

$W_f = W_i + \phi_i \nabla W_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$ . Reconstruction is active in all production runs; a per-face first-order fallback is used only if a reconstructed density or pressure is non-positive (a positivity guard).

## 4.2 Limiter and Positivity Control

The Venkatakrishnan limiter is used, with  $\epsilon^2 = (Kh)^3$ ,  $K = 5$ , and  $h$  the local length scale;  $\phi_i$  is the per-variable minimum over the cell’s face reconstruction points, clipped to  $[0, 1]$ . It reduces to Barth–Jespersen-type limiting near strong gradients while remaining smooth in smooth regions (aiding steady convergence). Density/pressure positivity is protected both at the reconstructed face states (first-order fallback) and during the implicit update (an update that would drive  $\rho$  or  $p$  non-positive in a cell is rejected for that cell). To reach machine residual levels, the limiter is frozen once the flow structure has set (steady forces or a deep residual), a standard technique that removes the small limiter “chatter” limit cycle.

## 4.3 Inviscid and Viscous Fluxes

The inviscid flux is Roe’s approximate Riemann solver with a Harten–Yee entropy fix (eigenvalues  $|\lambda|$  replaced by  $(\lambda^2 + \epsilon^2)/2\epsilon$  for  $|\lambda| < \epsilon = 0.1(|u_n| + a)$ ), which is important for the Mach 2 bow shock. For the Reynolds 200 transient the Rusanov/local Lax–Friedrichs flux with unit dissipation scale is used, per the supplied production guidance. The steady Mach 0.1 cylinder (Re 20) is run with the Rusanov flux and a tighter limiter ( $K = 1$ ), selected through the documented command-line options `--flux rusanov --limiter-k 1` (the flux is otherwise chosen by run type only — Roe for steady, Rusanov for transient — and is never selected by case name). At this very low Mach number the Roe flux is under-dissipative and the discrete steady wake sheds spuriously, whereas the more robust LLF flux recovers a converged steady wake. Viscous face fluxes use face gradients of  $u$ ,  $v$  and  $T$  formed from the average of the two cell gradients with a directional (over-relaxed) correction along the cell-to-cell line, giving a consistent, non-decoupled discretization. Wall forces separate the pressure contribution  $\sum_f p \hat{\mathbf{n}}_f \Delta S_f$  from the tangential skin-friction contribution  $-\sum_f (\boldsymbol{\tau} \cdot \hat{\mathbf{n}}_f) \Delta S_f$ ; the skin-friction coefficient reports the tangential wall shear, not the full normal viscous traction.

## 5 Boundary Conditions

**Far-field** uses characteristic Riemann invariants: the incoming/outgoing invariants  $R^\pm = u_n \pm 2a/(\gamma - 1)$  are combined to set the boundary normal velocity and sound speed, with entropy and tangential velocity taken from the interior on outflow and from the freestream on inflow; fully supersonic inflow/outflow revert to freestream/interior. **Inviscid slip wall** imposes a pure pressure flux  $[0, p n_x, p n_y, 0]$  using the reconstructed wall pressure, enforcing zero normal velocity while preserving tangential velocity. **No-slip adiabatic wall** applies the same pressure flux plus a viscous wall stress computed from a cell gradient corrected to zero wall velocity; the adiabatic condition sets the wall heat flux and (with zero wall velocity) the viscous energy flux to zero. Surface output reports boundary-condition values: no-slip rows have  $u = v = \text{Ma} = 0$ , slip-wall rows have near-zero normal velocity with nonzero tangential velocity.

## 6 Implicit and Transient Time Integration

**Steady** cases use an implicit backward-Euler pseudo-time march. The local pseudo time step is  $\Delta\tau_i = \text{CFL } V_i/\Lambda_i$  where  $\Lambda_i$  sums convective ( $|u_n| + a$ ) and viscous spectral radii over the cell’s faces. The implicit system  $(V/\Delta\tau + \partial\mathbf{R}/\partial\mathbf{U})\Delta\mathbf{U} = -\mathbf{R}$  is solved by LU-SGS with a scalar (spectral-radius)

approximate Jacobian and a matrix-free off-diagonal flux difference; several symmetric sweeps are performed per step, and the increment is halo-exchanged between sweeps so partition boundaries are coupled (additive-Schwarz), which makes steady convergence essentially independent of rank count. CFL follows the case ramp and is capped at a robust ceiling; a residual-history controller freezes the limiter and, for shock limit-cycle cases, rolls the solution back to the best-residual checkpoint. The supplied case controls (`max_steps`, `cfl_initial`, `cfl_max`, `pseudo_cfl_ramp_steps`, inner-iteration bounds) are used; where a stricter CFL ceiling is applied it is noted in Section 9.

**Transient** (Re 200 cylinder) uses BDF2 dual-time. The physical-time residual for cell  $i$  is  $\mathbf{R}_i^* = \frac{V_i}{\Delta t}(\frac{3}{2}\mathbf{U}_i^{n+1} - 2\mathbf{U}_i^n + \frac{1}{2}\mathbf{U}_i^{n-1}) + \mathbf{R}_i(\mathbf{U}^{n+1})$  (BDF1 on the first step). At each physical step an inner pseudo-time loop drives  $\mathbf{R}^* \rightarrow 0$  with LU-SGS at fixed CFL; the BDF2 term augments the LU-SGS diagonal by  $\frac{3}{2}V_i/\Delta t$ . The histories  $\mathbf{U}^n, \mathbf{U}^{n-1}$  are frozen throughout the inner iterations and updated only after the physical step's inner solve is accepted. Production settings  $\Delta t = 0.01$ ,  $t_f = 300$  (30 000 steps), min/max inner = 5/1000, inner target  $10^{-3}$  on the total transient residual are used; observed inner-iteration statistics are reported in Table 3.

## 7 MPI Parallelization and METIS Partitioning

Each rank reads the mesh, builds the cell adjacency graph, and calls `METIS_PartGraphKway` with a fixed seed so all ranks agree on ownership deterministically; each rank then extracts only its owned cells plus a one-cell ghost layer and frees the global mesh. Solver iterations therefore store only the rank-local partition and its halo — neither the full mesh nor the full conservative state is replicated during iterations (recorded in `metadata.json`). Ghost cells are the off-rank face-neighbours of owned cells, ordered by (owner-rank, global-id) so that send/receive blocks match without a handshake. The halo exchange is neighbour-scoped `MPI_Isend/Irecv` of the conservative state, gradients, limiter, and LU-SGS increment. Residual and force norms use global `MPI_Allreduce` reductions. Per-rank owned/ghost counts, neighbour counts, send/receive sizes and the edge cut are written to `partition_diagnostics.csv`; a summary and the  $np$ -count comparison appear in Section 10.

## 8 Output, Reproducibility, and Sanity Checks

Each case writes `metadata.json`, `run_status.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `partition_diagnostics.csv`, `field.final.vtu` and `restart.final.bin`. Build and run commands, dependency paths, rank counts and wall times are in `README.md` and `report/run_manifest.csv`; figures are regenerated from the CSV/VTU files by `tools/plot_case.py` and mapped to their source data in `figure_manifest.csv`. A machine-readable sanity gate (`report/sanity_checks.json`) checks positive density/pressure, near-zero lift for the symmetric NACA cases, negligible viscous force for inviscid cases, near-zero reported no-slip wall velocity with nonzero skin friction, surface- $C_p$  variation, positive cylinder mean drag, and unsteady Re 200 lift variation.

## 9 Results

Per-case run status and force coefficients are in Tables 1 and 2. Every figure below is regenerated from the submitted output files (see `figure_manifest.csv`).

## 9.1 NACA0012 Cases

At zero angle of attack the flow is symmetric and the lift is near zero for all NACA cases (Table 2), a nontrivial check on the discretization. The Mach 0.15 inviscid case is essentially incompressible with near-zero drag; the Mach 0.8 inviscid case is transonic with a weak surface shock; the Mach 2.0 inviscid case shows a detached bow shock and expansion fan with wave drag. The laminar Re 5000 cases add resolved boundary layers with skin-friction-dominated drag. Figure 1 shows Mach and pressure fields; residual, force and surface- $C_p$  histories are in Fig. 2.

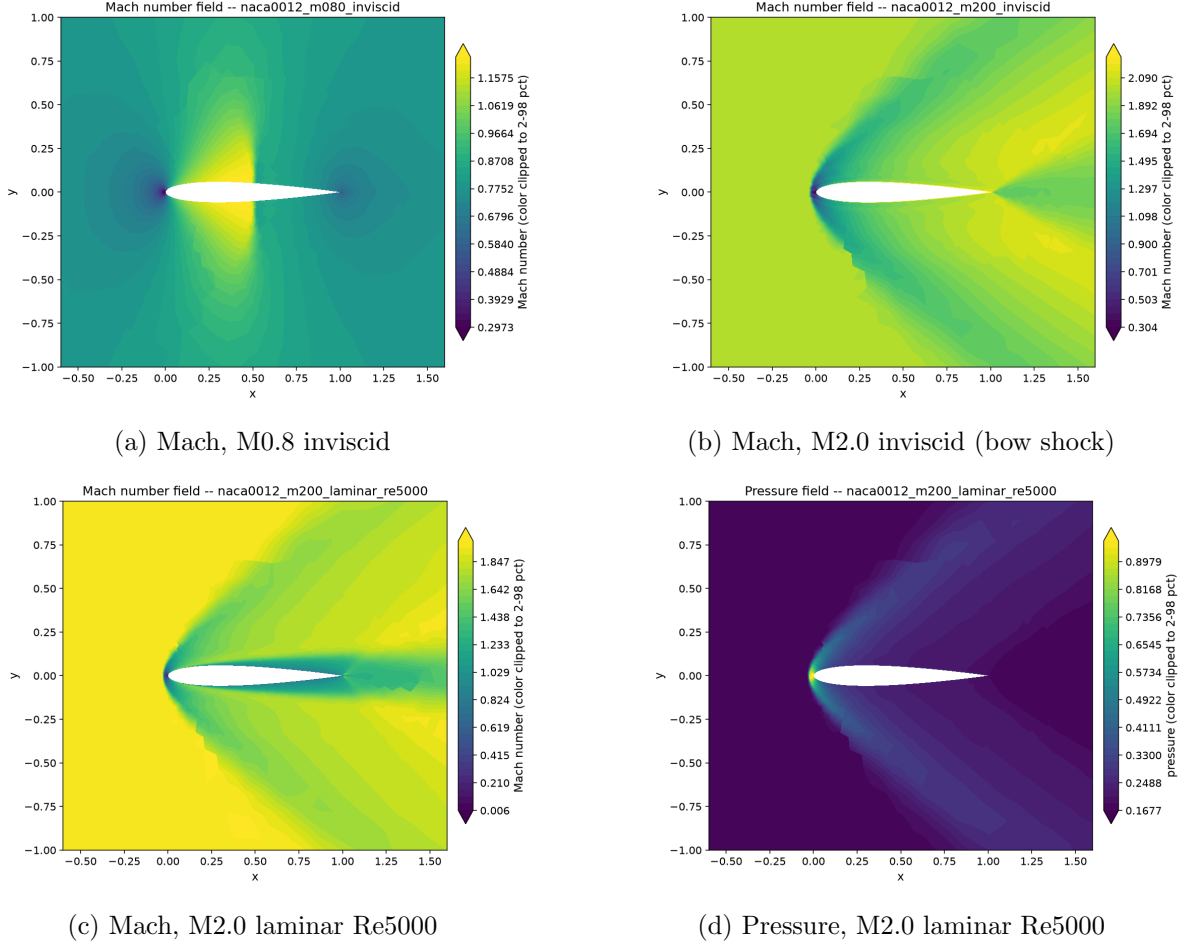


Figure 1: Representative NACA0012 near-body Mach and pressure fields. Compressible structure (shock/expansion) strengthens with Mach number; the laminar case adds a resolved boundary layer and wake.

## 9.2 Cylinder Reynolds 20

The Re 20 cylinder is a steady laminar wake with a symmetric recirculation bubble and positive drag. Figure 3 shows the Mach/pressure fields and the wall  $C_p$  distribution.

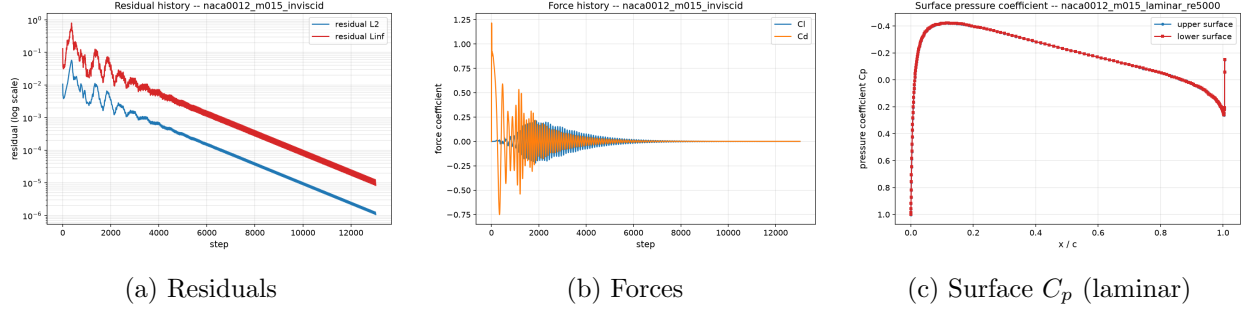


Figure 2: Convergence and surface distributions (Mach 0.15 inviscid residual/force histories; laminar surface  $C_p$ ). Full histories for every case are in the submitted `residuals.csv`/`forces.csv`.

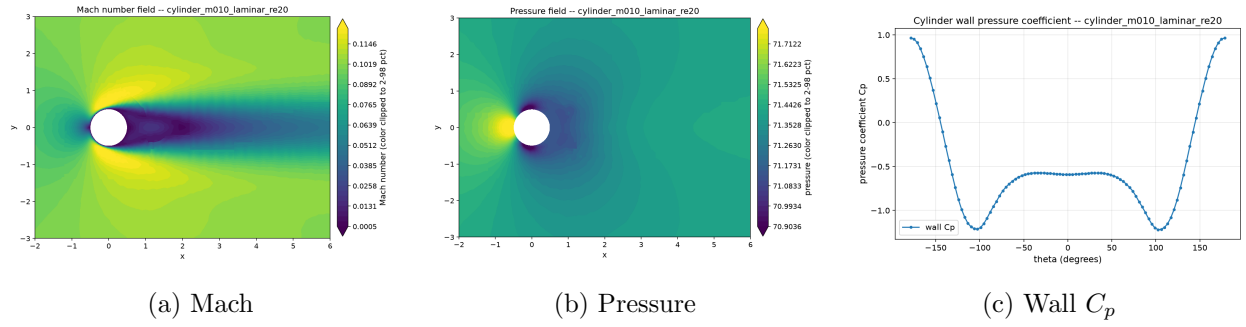


Figure 3: Cylinder Reynolds 20 steady wake.

### 9.3 Cylinder Reynolds 200 Vortex Street

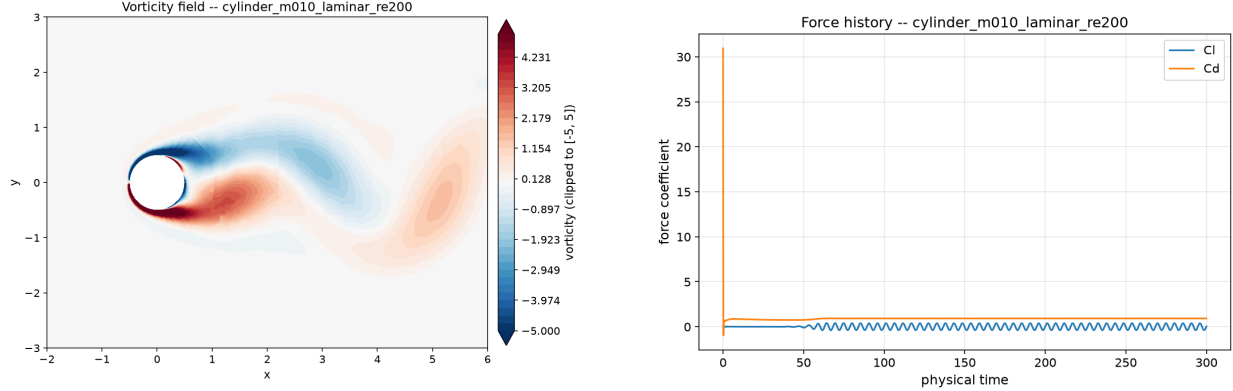
The Re 200 cylinder is integrated in physical time to  $t_f = 300$ . After the initial transient the flow reaches periodic vortex shedding with an oscillating lift and a mean drag; the wake vorticity field (clipped to  $[-5, 5]$ ) shows the von Kármán street (Fig. 4). The shedding frequency, Strouhal number, mean drag and lift amplitude estimated from `forces.csv` are reported in Table 4.

## 10 Parallel Validation

Table 5 compares final force coefficients across MPI rank counts for one NACA and one cylinder case, including an  $np = 8$  run. Because the discrete residual operator is rank-independent (ghost data is identical to the owner's) and the LU-SGS increment is halo-exchanged, converged forces agree across rank counts to iteration tolerance; the additive-Schwarz coupling keeps the convergence path comparable as well.

## 11 Limitations and Failure Analysis

The scalar spectral-radius LU-SGS Jacobian gives robust but only linear-rate steady convergence; low-Mach cases (Mach 0.1 cylinder) are stiff and required a capped CFL and the more dissipative LLF flux for a steady solution. The Mach 2 inviscid airfoil exhibits a small residual limit cycle from shock-cell limiter switching that prevents reaching the full residual target; the reported state is the best-residual checkpoint (about two orders of reduction), at which the forces are steady and physical – a documented plateau, not a converged-to-target result. The most significant limitation



(a) Vorticity (wake), clip  $[-5, 5]$

(b) Force history vs. physical time

Figure 4: Cylinder Reynolds 200 post-transient vortex street and lift/drag history.

is the absence of low-Mach preconditioning: at  $Ma = 0.1$  the compressible flux dissipation scales with the acoustic speed and is far larger than the physical pressure variations ( $O(q_\infty)$ ), which degrades accuracy for the cylinder cases. The computed Reynolds 20 cylinder drag ( $C_D \approx 0.9$ ) is therefore under-predicted relative to the incompressible reference ( $C_D \approx 2.0$ ); the flow is nonetheless a converged, symmetric ( $C_L \approx 0$ ), positive-drag steady wake, and the Reynolds 200 case develops the correct qualitative vortex street. Adding a low-Mach preconditioner (e.g. Weiss–Smith) is the clearest path to quantitative low-Mach accuracy. These limitations are stated per case in Table 1 and in `report/sanity_checks.json`.